

Tekken 3 Inspired Fighting Game - Complete Development Roadmap

From Scratch to Final Deployment

This roadmap covers the complete lifecycle of building a modern Tekken-style fighting game, including 2D prototype development, full 3D HD production, character systems, combat engine, multiplayer backend, cloud infrastructure, testing, deployment, monetization, analytics, and live operations.

Phase 1: Pre-Production

• Game Design Document (GDD) • Character roster planning • Combat mechanics • Story mode and game modes • UI/UX planning • Technical architecture

Phase 2: Technology Stack

Engine: Unity 6 (or Unreal Engine 5) Frontend: C#, UI Toolkit, Cinemachine Backend: Node.js, Express, PostgreSQL Cloud: Google Cloud, Firebase, Cloud Storage

Phase 3: 2D Prototype

Movement, jumping, blocking, health system, hitboxes, hurtboxes, combo detection, state machine, training mode.

Phase 4: Combat Engine

Idle, Walk, Run, Jump, Attack, Block, Knockdown, Victory states. Frame data system, combo engine, input buffering.

Phase 5: Animation Pipeline

Blender rigging, skeleton creation, IK setup, combat animations, throws, finishers and victory poses.

Phase 6: Full 3D HD Version

High-poly sculpting, game-ready models, PBR textures, HDR lighting, bloom, SSAO, volumetric effects.

Phase 7: Arenas

Urban, Temple, Desert, Snow and Volcano arenas with dynamic environmental effects.

Phase 8: AI Opponents

Beginner AI, Behavior Trees, ML-based adaptive AI.

Phase 9: Audio

Voice acting, combat SFX, dynamic battle music.

Phase 10: Multiplayer Backend

Ranked matchmaking, casual matches, tournament mode, authoritative server architecture.

Phase 11: Database

User profiles, match history, rankings, cosmetics.

Phase 12: Monetization

Skins, battle pass, cosmetic store, DLC fighters.

Phase 13: Analytics

Firebase Analytics, BigQuery, player retention, gameplay balancing and revenue tracking.

Phase 14: Testing

Unit tests, multiplayer tests, performance optimization. Target: 60 FPS Mobile, 120 FPS PC.

Phase 15: CI/CD

GitHub Actions, automated builds, deployment pipelines.

Phase 16: Deployment

Android, iOS, Steam and Console deployment.

Estimated Timeline

MVP: 6-8 Months Full Production Game: 12-18 Months Team Size: 6-8 members